

Electric Vehicles (EE4052)

Final Exam

Date: December 19, 2025

Course Instructor

Dr. Muhammad Naeem

Total Time (Hrs): 3

Total Marks: 50

Total Questions: 6

Roll No

Section

Student Signature

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Attempt all the questions.

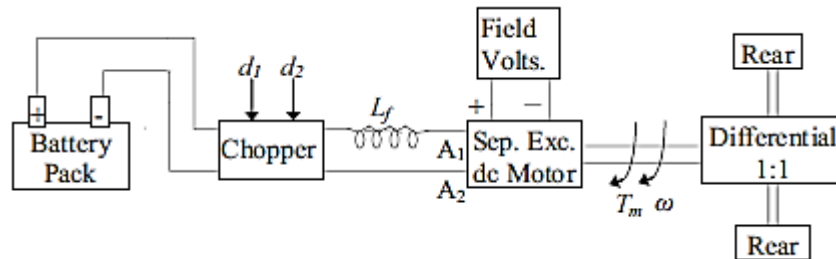
Sketch the waveforms on the question paper and return the paper with the answer sheet.

Think wisely before attempting any question. Read the statement twice and then start with clear mindset. Avoid cutting and overwriting.

Formulae are available at the last page.

CLO 1: Apply the principles of electric vehicle architectures, propulsion systems, and energy storage technologies.

Q1: Consider the electric vehicle drivetrain driven by a two-quadrant chopper shown in figure below. d_1 is the duty ratio for the acceleration operation, while d_2 is the duty ratio for braking operation. The various parameters are given below: **[10 Marks]**



The EV parameters are:

$\rho = 1.16 \text{ kg/m}^3$, $m = 850 \text{ kg}$, $C_D = 0.2$, $A_F = 2 \text{ m}^2$, $g = 9.81 \text{ m/s}^2$, $C_o = 0.009$ and $C_1 = 0$, $r_{wh} = 0.28 \text{ m}$
Motor and controller parameters are:

$R_a = 0.1 \Omega$, $L_a = 2 \text{ mH}$, $k\phi = 0.6 \text{ Vs}$, $I_{a, \text{rated}} = 200 \text{ A}$, $f_s = 500 \text{ Hz}$, $L_f = 1.6 \text{ mH}$

Currently the vehicle is moving on a level road with constant speed of 25 m/s while the chopper is being switched as $d_1 = 0.4$ and $d_2 = 0$.

- Determine** the steady-state power P_{TR} being drawn from the battery.
- Determine** whether the steady-state operation of the chopper is in CCM, DCM or UNCM.
- Determine** the steady-state operating point of the vehicle (ω and T_{TR})
- Calculate** the percent ripple in the armature current for the operating point calculated in part c.
- Calculate** the maximum percent grade of the vehicle in this operating state.

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CLO 2: Analyze the performance and efficiency of electric powertrains and energy storage systems.

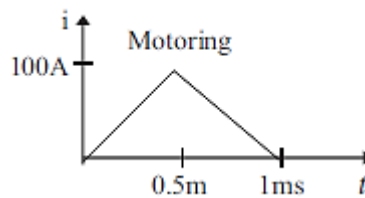
Q2: Given below are constant discharge characteristics of a 12 V, 100 Ah lead acid battery: **[10 Marks]**

SP (W/kg)	SE (Wh/kg)
10	67.5
110	8

The battery characteristics are to be expressed in terms of Peukert's equation of the form:

$$(SP)^n(SE) = \lambda$$

- Calculate** the constants n and λ , assuming a linear relationship between $\log(SP)$ and $\log(SE)$.
- Find** the capacity Q_T of the battery if the theoretical energy density is $SE_T = 67.5$ Wh/kg, given that battery mass of 15 kg.
- Consider a battery pack consists of four parallel sets of six series connected battery units having above characteristics. One steady-state motoring (discharge) cycle of battery pack current is shown in figure below. **Calculate** the time it takes to reach 80% DoD.



CLO 3: Design electric drives, battery management systems and power electronic converters to enhance vehicle performance and efficiency.

Q3: Design a three-phase induction motor drive system rated at 10 hp, 200 V, 60 Hz, 6-pole, wye-connected, suitable for use in an electric vehicle traction application. The rated speed of the motor is 1164 rpm. Other parameters are as follows: **[10 Marks]**

$R_s = 0.28 \Omega$	$L_s = 0.055$ H	$L_m = 0.054$ H
$R_r = 0.18 \Omega$	$L_r = 0.056$ H	Turns ratio = 2

- Formulate** the dq -axis equivalent model of the motor by transforming the stator and rotor voltages into the synchronous reference frame considering the motor is stalled.
- Solve** the equivalent model formulated in part a) for the stator and rotor currents
- Solve** the equivalent model of part a) for the output electromechanical power P_e
- Solve** the equivalent model for the electromagnetic torque T_e

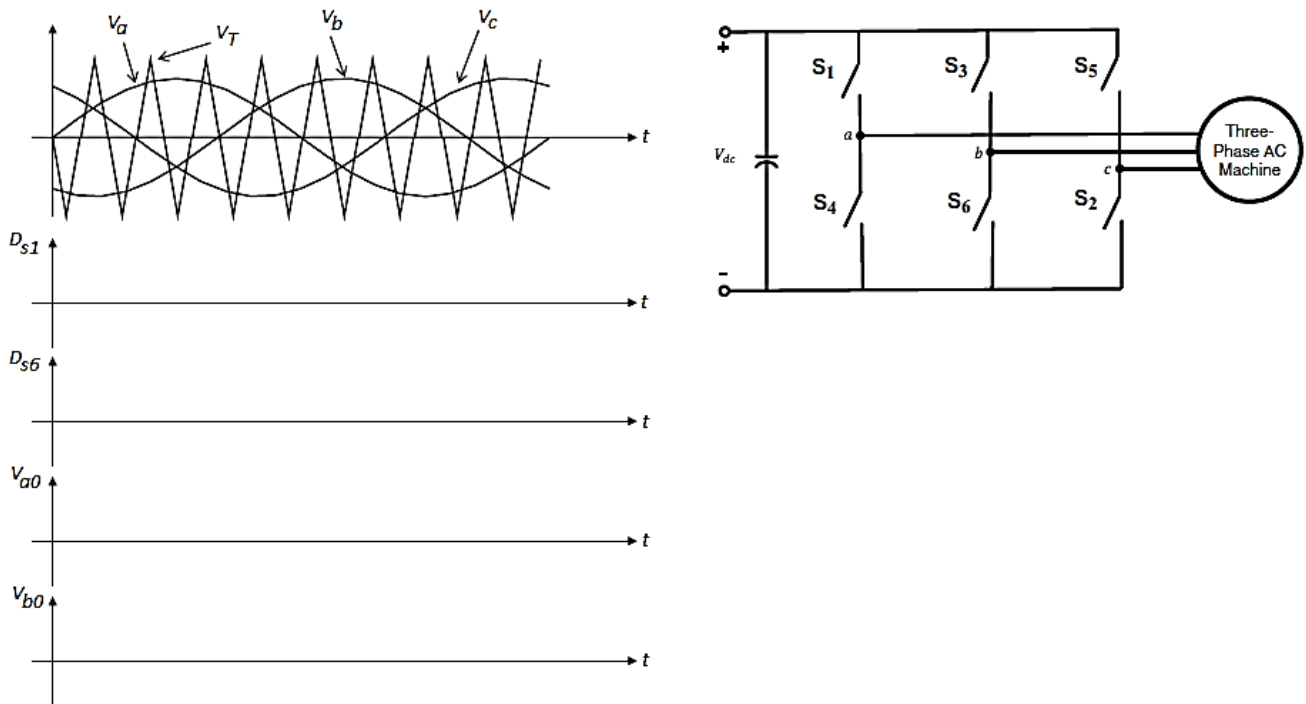
CLO 3: Design electric drives, battery management systems and power electronic converters to enhance vehicle performance and efficiency.

Q4: A three-phase SPWM-controlled traction inverter is to be designed for the EV powertrain.

- Generate** the switching signals for S_1 and S_6 on the timing diagram shown below. Also sketch the phase voltages V_{a0} and V_{b0} on the diagram shown on next page. (Sketch on question paper and attach it with the answer sheet).
- Design** the SPWM parameters (magnitude and frequency of all voltages) such that its switching frequency is 10 kHz, the amplitude modulation index is 0.9 and the frequency modulation index is 200. **[5 Marks]**

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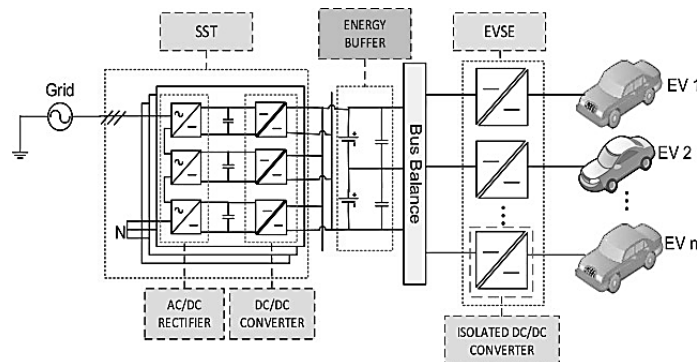


CLO 3: Design electric drives, battery management systems and power electronic converters to enhance vehicle performance and efficiency.

Q5: Design a DC-DC boost converter using an ideal switch for the DC machine drive that takes 24-V fixed voltage from a DC bus and converts it to 42-V. Based on the designed Duty cycle D , plot the inductor voltage V_L , inductor current I_L , diode current i_L , average load current i_o , and load voltage v_o when the load resistor is $1\ \Omega$. **[5 marks]**

CLO 4: Analyze the grid interfacing and charging infrastructure for electric vehicles and figure out its impacts on power grid.

Q6: Analyze the overall design architecture, ratings, and layout of an Ultra-Fast Charging (UFC) station that is built with a front-end MV SST connected to 11 kV MVAC grid with an allowable voltage variation of $\pm 5\%$. The modular topology to be used is shown in figure below. The front-end power converter is built with a multi-level topology of cascaded H-bridges consisting of eight three-phase modules with each module having three single-phase sub-modules. Each sub-module consists of an AC/DC front stage and an isolated DC/DC stage to establish the LVDC bus voltage level for Electric Vehicle Supply Equipment (EVSE) that serves as interface between EVs and charging station. The rated input power of the charging station is 1 MW and it is capable of operating at 0.9 lag/lead power factor. The average power demand of the charging station is 50% of the rated power while its utilization factor at rated power per day is 15%. **[10 Marks]**



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- Calculate** the maximum reactive power that UFC is capable of sourcing/sinking on MVAC side.
- Calculate** the rated AC voltage level for each front-end AC/DC stage (sub-module)
- Calculate** the rated current for each front-end AC/DC stage (sub-module) (Assuming unity power factor)
- Calculate** the energy requirement of the charging station per day and the average power demand.
- It is planned to install a PV-based Distributed Generator (DG) and an Energy Storage System (ESS) of combined rated capacity of 800 kWh at the UFC, **calculate** how much energy would be drawn per day from the grid after the installation.

Formulae:

$$F_{roll} = mg \cos \beta (C_o + C_1 v^2); \quad F_{AD} = \frac{\rho}{2} C_D A_F v^2; \quad F_{gXT} = mg \sin \beta; \quad \tan \beta = \frac{F_{TR}}{\sqrt{(mg)^2 - F_{TR}^2}}$$

$$K_1 = \frac{F_{TR}}{m} - gC_0; \quad K_2 = \frac{\rho}{2m} C_D A_F + gC_1; \quad V(t) = \sqrt{\frac{K_1}{K_2}} \tanh \sqrt{K_1 K_2} t$$

$$s(t) = \frac{1}{K_2} \ln (\cosh(K_2 V_T t)); \quad t_f = \frac{1}{\sqrt{K_1 K_2}} \tanh^{-1} \left(\sqrt{\frac{K_2}{K_1}} V \right)$$

$$\begin{bmatrix} v_{ds} \\ v_{qs} \\ v_{dr} \\ v_{qr} \end{bmatrix} = \begin{bmatrix} R_s & -\omega L_s & 0 & -\omega L_m \\ \omega L_s & R_s & \omega L_m & 0 \\ 0 & -(\omega - \omega_r) L_m & R_r & -(\omega - \omega_r) L_r \\ (\omega - \omega_r) L_m & 0 & (\omega - \omega_r) L_r & R_r \end{bmatrix} \begin{bmatrix} i_{ds} \\ i_{qs} \\ i_{dr} \\ i_{qr} \end{bmatrix} + \begin{bmatrix} L_s & 0 & L_m & 0 \\ 0 & L_s & 0 & L_m \\ L_m & 0 & L_r & 0 \\ 0 & L_m & 0 & L_r \end{bmatrix} \begin{bmatrix} p i_{ds} \\ p i_{qs} \\ p i_{dr} \\ p i_{qr} \end{bmatrix}$$

$$\begin{bmatrix} v_q \\ v_d \\ v_0 \end{bmatrix} = \frac{2}{3} \begin{bmatrix} \cos \theta & \cos(\theta - \frac{2\pi}{3}) & \cos(\theta + \frac{2\pi}{3}) \\ \sin \theta & \sin(\theta - \frac{2\pi}{3}) & \sin(\theta + \frac{2\pi}{3}) \\ 0.5 & 0.5 & 0.5 \end{bmatrix} \begin{bmatrix} v_a \\ v_b \\ v_c \end{bmatrix}; \quad T_e = \frac{3}{2} \frac{P}{2} L_m (i_{qs} i_{dr} - i_{ds} i_{qr})$$

$$\vec{v}_{qds} = R_s i_{qds} + L_s (p \vec{i}_{qds}) + L_m (p \vec{i}_{qdr}) + j\omega (L_s \vec{i}_{qds} + L_m \vec{i}_{qdr})$$

$$\vec{v}_{qdr} = R_r i_{qdr} + L_r (p \vec{i}_{qdr}) + L_m (p \vec{i}_{qds}) + j(\omega - \omega_r) (L_r \vec{i}_{qdr} + L_m \vec{i}_{qds})$$

$$\text{Area of Triangle} = \frac{1}{2} \times \text{base} \times \text{height}$$

CCM:

$$I_{a0} = \frac{V_{bat}}{R_a} \left(\frac{e^{\frac{dT}{\tau}} - 1}{\frac{T}{e^{\tau}} - 1} \right) - \frac{E}{R_a}; \quad I_{a1} = \frac{V_{bat}}{R_a} \left(\frac{1 - e^{\frac{-dT}{\tau}}}{1 - e^{\frac{-T}{\tau}}} \right) - \frac{E}{R_a}; \quad \langle \omega \rangle = \frac{dV_{bat}}{k\phi} - \frac{R_a}{(k\phi)^2} \langle T \rangle$$

DCM:

$$I_{a2} = \frac{V_{bat}}{R_a} \left(\frac{e^{\frac{dT}{\tau}} - 1}{\frac{T}{e^{\tau}} - 1} \right) - \frac{E}{R_a}; \quad I_{a1} = \frac{V_{bat}}{R_a} \left(\frac{1 - e^{\frac{-dT}{\tau}}}{1 - e^{\frac{-T}{\tau}}} \right) - \frac{E}{R_a}; \quad \langle \omega \rangle = \frac{d'V_{bat}}{k\phi} - \frac{R_a}{(k\phi)^2} \langle T \rangle$$

$$\lambda_{ds} = (L_{ls} + L_{md}) i_{ds}; \quad \lambda_{qs} = (L_{ls} + L_{mq}) i_{qs} - \lambda_f$$

$$P = i_{ph}^2 R_s + \frac{d}{dt} (L_{ph} i_{ph}^2) + \frac{1}{2} i_{ph}^2 \frac{dL_{ph}}{d\theta} \omega; \quad T = \frac{1}{2} i_{ph}^2 \frac{dL_{ph}}{d\theta} \omega$$